

Shivani Dwivedi

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Objective

Computer Science student with strong foundations in data structures, algorithms, operating systems, and computer networks, seeking a Software Engineering Internship to apply my skills in systems programming, networking, and software development while building scalable and secure technology solutions.

Education

- **Master Of Computer Applications** Present
National Institute of Technology, Bhopal | CGPA: 8.11
- **Bachelor of Computer Applications** 2022
Dr. Rammanohar Lohia Avadh University | 70.19%

Technical Skills

- **Programming Languages:** C, C++, Java, Python
- **Systems & Networking:** TCP/IP Socket Programming, POSIX, Multi-threading, RESP Protocol
- **Tools:** VS Code, Git, Linux, CMake, SDL2
- **Core Concepts:** Data Structures & Algorithms, Object Oriented Programming and Design, Operating Systems, Computer Networks, Database Management System (DBMS)

Projects

Chip8 Emulator | C++, SDL, CMake

[GitHub](#)

- Built a CHIP-8 emulator in C++ with support for all 35 standard instructions, 4KB memory, 16 general-purpose registers, stack management, and opcode decoding.
- Implemented XOR-based sprite rendering with collision detection on a 64×32 display using **SDL2 and low-level bitwise operations**.
- Added keypad input, delay and sound timers, and ROM loading to correctly execute standard CHIP-8 test ROMs and classic games.

In-Memory Key-Value Server (Redis-Inspired) | C++, TCP Sockets, CMake, RESP

[GitHub](#)

- Built a Redis-inspired in-memory key-value server in C++ using **TCP sockets** and the **RESP protocol** for client-server communication.
- Implemented core **Redis commands** (GET, SET, DEL) with **TTL-based key expiration**, request parsing, command dispatching, and response serialization.
- Designed a modular architecture separating networking, protocol handling, command execution, and storage, with support for concurrent client handling.

System Performance Monitoring & Anomaly Detection | Python, Scikit-learn, Pandas, NumPy

[GitHub](#)

- Built a real-time ML-based system monitoring tool for CPU/RAM metrics with **Random Forest ($R^2=0.98$)** and **LSTM ($R^2=0.74$)** prediction models.
- Developed an **Isolation Forest anomaly detector** with hybrid rule-based filtering, reducing false alerts by **~80%**.
- Implemented **process-level root cause analysis** using psutil, identifying suspicious processes with CPU, RAM, threads, and network usage details.

Achievements and Open Source

- **Leetcode: Peak Rating 1784 (Top 8.75 globally) | 525+ problems solved** across all difficulty levels | Achieved **Global Rank 1605** in **Weekly Contest 502**.
- **rs-stl (Rust STL Implementation):** Implemented **sort_heap** algorithm for rs-stl — a Rust port of C++ STL — with full maintainer code review.

Position Of Responsibility

Technical Head | Matrix Club, MANIT Bhopal

- Organized technical workshops, coding events, and club activities and mentored members in C++, DSA.